

Impact of local timescales in a cellular automata model of excitable media

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Abstract

A cellular automata system known as the Greenberg-Hastings cellular automata (GHCA), with cyclic and excitable individual dynamics, provides a useful paradigm for modeling emergent phenomena in an excitable medium. Most past studies of GHCA have investigated the asymptotic dynamics of the system, assuming fixed extents for the active and passive dynamical phases. This study is a systematic investigation of the relationship between the dynamical properties of the individual cells of a small system implementing excitable dynamics, and the emergent system wide asymptotic state. The impact of the temporal extent of the individual dynamical phases (τ_a, τ_p) , on the probability of persistence $P(\tau_a, \tau_p)$, defined as the fraction of initial configurations which sustain spatiotemporal oscillations asymptotically, is examined for each unique configuration of the system. The main result obtained is that the probability of persistence $P(\tau_a, \tau_p)$ is found to assume a characteristic sigmoidal form, as the dynamical behaviour of the system transitions from absolute quiescence to persistent spatiotemporal oscillations, with respect to the parameters (τ_a, τ_p) . A plot of the numerical findings in the (τ_a, τ_p) space provides a consolidated view of their impact on the emergent behaviour of the system. In conclusion, some implications of the methods and results presented in this work, to the study of complex systems, are discussed.

I. INTRODUCTION

Models of complex dynamical systems have garnered a lot of research attention in recent times, owing to the ubiquity of their applicability [1]. The complex systems paradigm has proved fruitful in a wide range of disciplines, such as socioeconomic phenomena [2, 3], chemical reaction networks [4], epidemic spreading processes [5, 6], and many other such phenomena, which require an interdisciplinary approach for modeling and analysis. The role of computational modeling in understanding complex systems has steadily risen with growing computational capacity. In parallel, cellular automata (CA) systems have become established as a suitable framework for the general purpose of modeling spatially extended complex systems [7]. Applications of CA involve a wide range of phenomena, from excitable media [8] to the spread of infectious diseases [9], along with interesting applications in theoretical computer science and mathematics [10–12].

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Cellular automata systems are based on a discrete grid/lattice of cells, which may naturally represent spatial extent in two dimensions. Each cell within the lattice evolves from one discrete state to another, in discrete time, according to specified rules. The transition rules are applied locally, as a function of the current state of the cell as well as the states of the neighbouring cells. A CA system is well defined when the state space and the local dynamical rules are unambiguously specified.

Various approaches have been developed in order to analyse CA systems and their long term behaviour. Mathematical approaches have utilized frameworks developed with concepts of algebra [13], probability and measure theory [14] as well as topology and symbolic dynamics [15].

Approaches from the field of dynamical systems have involved analysis of attractors and their basin structures, validated by extensive computational simulation for specific systems, namely elementary CA [16, 17]. Concepts such as Lyapunov exponents [18, 19] have been utilized for characterization of asymptotic behaviours of such systems as well.

A seminal study in 1978, by J. M. Greenberg and S. P. Hastings [20], introduced a cellular automata model with three-phased, cyclic and excitable dynamics for modeling excitable media. The local dynamical properties of such an excitable medium correspond to the active and passive phase extents τ_a and τ_p respectively, defined for each individual cell in a lattice. Along with other approaches to model and analyze excitable media [21, 22], the Greenberg-Hastings CA model (GHCA model) has proven itself to be a fruitful framework. For instance, the GHCA model was found to be a suitable approach to model Belousov-Zhabotinsky reactions, which are a prototypical example of excitable media [23], and it continues to reveal insights about such systems [24]. In general, GHCA has been used to model processes which typically involve threshold based spreading of “*activity*” via local contact. It belongs to a broader class of CA with cyclic, excitable local dynamics [25]. The cyclic local dynamics, along with associated receptive, active and passive phases, naturally emulate the observed behaviour of a general excitable medium. Computational models with such features have been implemented for modeling biological phenomena, such as spatiotemporal patterns of cardiac rhythms [26, 27], macroscopic brain waves [28], as well as in the context of epidemic processes [29, 30].

In the model originally proposed by Greenberg and Hastings [20], the active and passive phases were assigned a single integer state each. Therefore the state space of an individual

cell was $S = \{0, 1, 2\}$, where $\{0\}$ was the receptive phase, $\{1\}$ was the active phase and $\{2\}$, the passive phase. It has been observed that such systems asymptotically approach either a steady state of spatiotemporal oscillations, which may be interpreted as a discrete analogue of *limit cycles*, or they approach a *fixed point*, devoid of activity in the system. Such asymptotic behaviours have been observed in related contexts also, such as mathematical models of biological control systems [31].

The spatial features of a 2D lattice configuration which lead to persistent oscillations in the two dimensional GHCA system, were explored by the original authors in a ensuing study [32]. A generalized version of such three-phased cyclic, excitable dynamics, where the active and the passive phases are each extended over a series of integer states $[1 \dots \tau_a]$ and $[\tau_a + 1 \dots \tau_a + \tau_p]$ respectively, was implemented, with the constraint $1 \leq \tau_a \leq \frac{\tau_a + \tau_p}{2}$, implying that the active phase extent is less than or equal to the passive phase extent. The study approached the question of persistent oscillations in the system, by investigating the spatial arrangement of individual states in an arrangement referred to as "continuous cycles", and calculating the "winding number" of such continuous cycles. A "cycle" is defined as a tuple of adjacent cells such that the last cell is adjacent to the first one. A direct implication of the conclusions of the study, specifically theorem 4, is that the presence of structural cycles is necessary for the persistence of activity in the system.

Now, a 2D lattice obviously consists of structural cycles, but it is not necessary in general. For instance, consider a tree graph, where each node has a state, and the edges define connections between nodes, such that each node has a well defined neighbourhood. Consider the Greenberg-Hastings dynamics implemented on such a graph. Since tree graphs are devoid of structural cycles by definition, observing persistence on such graphs should be impossible according to the implication described earlier, as long as the constraint $\tau_a \leq \tau_p$ is satisfied. This is quite easily verifiable by simulations, and was consistently observed. However, for longer active phases i.e. $\tau_a > \tau_p$, activity was found to persist even in tree graphs, i.e. in the absence of structural cycles. So it is clear that general conclusions regarding the system, with the restriction on the active phase extent, do not apply when the restriction is relaxed.

The depth of understanding of emergent oscillations in GHCA systems was significantly expanded in a subsequent study [14]. Focusing on the simplest state space, i.e. $S = \{0, 1, 2\}$, the study proves several important results, some of which hold for lattices of arbitrary dimensions, utilizing sophisticated methods from probability and measure theory. For infinite

lattices of dimension $d \geq 2$, three distinct asymptotic patterns are identified and analyzed, namely period 3, period 4 and extinction. The most noteworthy and general result is theorem 4, which presents limiting expressions for the probability of each of these behaviours, based on the rate of decay of the initial fraction of active and passive cells, as the lattice size tends to infinity. In the development of the proof, certain lemmas are presented as well. In the context of the present study, lemmas 5 and 6 are of specific relevance, as they provide conditions on the initial configuration, with specific focus on the smallest spatial sub-unit, the 2×2 box, which determine persistence or extinction of activity in the lattice. This provides the motivation to concentrate the present study, which considers arbitrary phase extents, on the 2×2 spatial unit of a 2D lattice. In the context of excitable media, this may be understood as a pattern/activity generating core.

That the question of phase extents is an important one, is further emphasised in a recent exhaustive study of the one dimensional version of the GHCA model [33] which considers arbitrary active and passive states on one dimensional infinite lattices. The study elucidates the clarity of understanding obtained by imposing such abstract structures upon the set of configurations of CA systems. In this detailed and rigorous study, the authors have decomposed the set of asymptotic configurations into subsets of configurations with qualitatively distinct behaviours (c.f. Theorem 4.5), which are referred to as "pulse collision and annihilation" and propagation of "stationary dislocations and defects", based on the dynamical patterns of the configurations which constitute the subsets. Several thought provoking structures and methods are used in the process. The local analysis of "2-blocks" to determine the conditions for change of "step sizes" (c.f. section 4.2) leads to some understanding of the distinction between the cases $\tau_a \leq \tau_p$ and $\tau_a > \tau_p$. As the authors of the study remark (c.f. remark 4.7), in one dimension, for the case of $\tau_a > \tau_p$, novel wave phenomena occur in the asymptotic state, such as sources emitting pulses to the left and right antisynchronously, which are qualitatively analogous to experimentally observed 1D spirals [34]. This motivates further investigation of such scenarios in two dimensional lattices, which are of specific interest as they provide a natural platform to model excitable dynamics in physically realizable contexts.

The GHCA model has also been studied on large 2D lattices, with statistical sampling of an ensemble of initial conditions, to investigate aspects such as the role of the initial proportions of receptive, active and passive phases [35], the role of synchronization of the initial

individual states [36], as well as the role of initial non-uniformity in the spatial distribution of states [37], in the emergence of persistent oscillations in the system. However, the studies were conducted for fixed active and passive phase extents, leaving the relationship between dynamical properties of the individual cells of the system, and the emergent system-wide oscillations, unexplored.

Generally, the impact of the extent and proportion of active and passive phases, which are temporal properties of the individual level dynamics, on the probability of persistent global spatiotemporal oscillations, is a relevant and relatively unexplored question, especially in the context of excitable media [24]. For instance, what is the nature of the transition in probability of persistence with increasing active or passive phase extent? Does a longer active phase imply certainty of persistent activity, or is there a gradual change in the persistence probability, as the system transitions from quiescence to persistence? These are the kind of questions that are explored in this study.

The primary result which is obtained in this study, is that the variation of the probability as a function of the phase extents takes on a characteristic sigmoidal form, as the system transitions from quiescence to persistence. The detailed numerical findings are consolidated into a parameter space plot, which provides further insights into the relationship between the individual level dynamical properties to the emergent global, asymptotic fate of the system.

The paper is organized as follows. In Section II a detailed description of the model, including the state space and transition rules, is presented. In Section III, a general framework to uniquely identify global configurations of a lattice is introduced. The framework is quite general and can be applied to any CA system irrespective of the kind of local state transition rules implemented. Numerical simulations are carried out for each initial condition within the set of all configurations of the system, with specified dynamical parameters (τ_a, τ_p) , and configurations which lead to persistent activity in the system are determined. This protocol is performed as the parameters are systematically varied. The general approach of exhaustively exploring the phase space of a CA has been implemented before (for e.g. see section 3.c in reference [16]), and is selected as the method for the present study. The observations and the parameter space plot, are presented in Section IV. A brief discussion in Section V summarises the study, and further directions are considered.

II. MODEL

Cells are the dynamical units which collectively form the CA system. To model the excitable dynamics of an individual cell, three dynamical phases are defined - *receptive*, *active* and *passive*. Each phase consists of an associated series of integer states. The receptive phase consists of a single state, namely $\{0\}$. A cell being in the receptive phase implies that it is ready to be excited by an active neighbouring cell. A receptive cell will remain receptive until activated, so the receptive phase may be considered as an absorbing phase. A cell in the active phase will activate each receptive cell in its neighbourhood. A passive cell is conferred temporary immunity, i.e. it neither affects, nor is affected by, any cell in its neighbourhood, and just proceeds through the associated sequence of states, and finally returns to the receptive phase. The local dynamics is cyclical, i.e. each cell, once excited/activated, goes through the entire sequence of states, transitioning from the active phase to the passive phase, and ultimately returns to the receptive phase.

As mentioned earlier, the model as originally proposed by Greenberg and Hastings assigned singular states for the active and passive phases. The state space was defined as $S = \{0, 1, 2\}$, where $\{0\}$ was the receptive phase, $\{1\}$ was the active phase and $\{2\}$, the passive phase. The generalized version of such three-phased dynamics, where the active and the passive phases are each extended over a series of integer states, is considered here. The extent of the dynamical cycle is denoted as $\tau_a + \tau_p = \tau_0$, where τ_a is the extent of the active phase and τ_p is the extent of the passive phase. Consequently, the state space of each individual cell in the system is defined as $S = \{0, 1 \dots \tau_a, \tau_a + 1 \dots \tau_0\}$, with $|S| = K = \tau_0 + 1$ states. K is referred to as the *cycle duration*.

The transition rules which describe the individual dynamics are as follows. A receptive cell, when excited by a neighbouring active cell, will enter the active phase which extends from $\{1 \dots \tau_a\}$, then enter the passive phase which follows from $\{\tau_a + 1 \dots \tau_0\}$, subsequently returning to the receptive phase $\{0\}$. The receptive phase is an absorbing phase, and is assigned a singular state $\{0\}$. Since τ_a and τ_p represent the extent of the active and passive dynamical phases respectively, it is implied that $\tau_a, \tau_p > 0$. Hence, the two key parameters in our study are the extents of the active and passive phases, τ_a and τ_p respectively, which specify the local dynamics occurring at each individual site.

An illustration of the evolution of the global configuration of a 5×5 lattice over a single

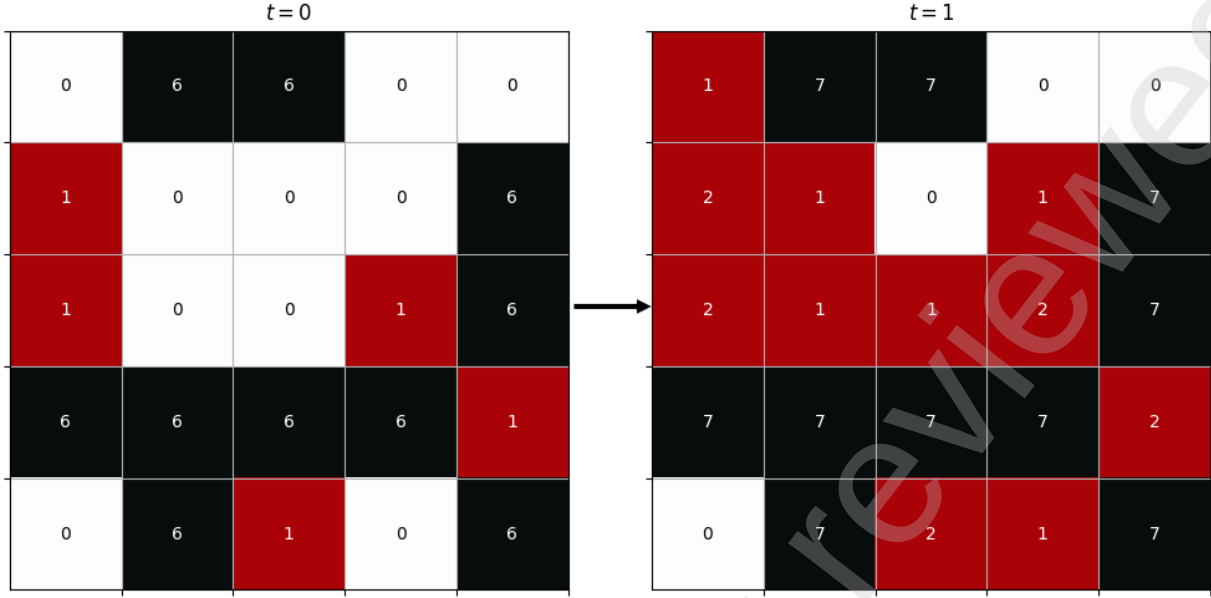


FIG. 1. Consecutive states of the lattice. Linear extent $L = 5$, number of cells $N = 5 \times 5$, active and passive phases $\tau_a = \tau_p = 5$, cycle duration $K = 11$. Colours represent phases — white - receptive; red - active; black - passive.

time step is shown in Fig. 1. All cells are updated simultaneously. The cells which are initially active or passive, increment their states by 1, i.e. $0 < \tau_{i,j}^t \longrightarrow \tau_{i,j}^t + 1 \pmod{K}$. Since the active and passive phases are extended over a series of states, i.e. $\tau_a, \tau_p > 1$, they may remain in the same phase or shift to the subsequent phase, which is indicated by the colour of the cell. However, the receptive cells which are initially in the neighbourhood of active cells, switch to the active phase. In subsequent steps, the activity spreads to nearby receptive cells, while simultaneously the passive cells reach the end of their phase and return to being receptive. The interplay of such deterministic, cyclic, three-phased dynamics at the local level, with the initial spatial distribution of the states, leads to the asymptotic regimes of either self-sustained oscillations or cessation of activity in the system.

Having understood the basic rules of time evolution which define the cellular automata model, a procedure for exhaustively parsing the space of initial configurations of the system is developed in the next section. Note that a system is well defined when the dimensions $N = L \times L$, and the dynamics, defined by the parameters (τ_a, τ_p) , are specified. Enumerating each configuration of a well defined system uniquely, will be the first step towards systematically studying the probabilities associated with the various parameter regimes of the system.

III. METHODS

In order to calculate the probability of a given initial configuration evolving to either the steady state with persistent activity, or the quiescent state with no activity, let us begin by describing the quantity which will form the backbone of the entire study, the *probability of persistence*, or $P(\tau_a, \tau_p)$. This is the fraction of the total number of configurations, which asymptotically retain persistent activity. For a given set of parameters L and K , all possible configurations of $N = L^2$ cells, arranged in a square grid with linear extent L , and a state space S with K possible states for each cell, are enumerated. The dynamics of the system is defined by (τ_a, τ_p) , such that $K = |S| = \tau_a + \tau_p + 1$. The total number of configurations can be calculated for a defined system, and after computationally confirming which asymptotic state each configuration evolves to, the probability of persistence $P(\tau_a, \tau_p)$ can be ascertained. In the following subsections, we describe the method in detail.

A. Uniquely identifying global configurations of a lattice

We illustrate the process of identifying each configuration uniquely, for the relatively accessible case of $N = 4$ cells, arranged in a 2×2 grid, with $K = 3$ states ($S = \{0, 1, 2\}$, $\tau_a = \tau_p = 1$). Since each of the N cells can be in any one of the K states at any given time, we can conclude in a straightforward manner that there are $K^N = 3^4 = 81$ configurations of the given system.

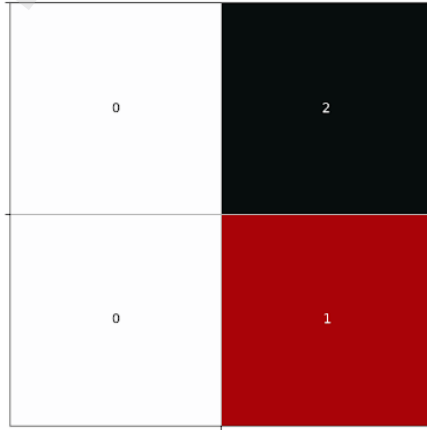


FIG. 2. A specific configuration of the system. $N = 2 \times 2$, $(\tau_a, \tau_p) = (1, 1)$ (cf. text for definitions of N, τ_a and τ_p).

Fig. 2 represents one of the 81 possible configurations of the system. Considering each cell as an independent dimension, the configuration in Fig. 2 may be specified as $(0, 2, 0, 1)$, in the conventional row major order of cells. It should be observed that since the state space has $K = 3$ symbols, the configuration $(0, 2, 0, 1)$ of $N = 4$ cells, can be thought of as $0201_{(base\ 3)}$, which is a 4 digit number in *base 3* notation. In order to uniquely identify this specific configuration, it is sufficient to convert this 3-ary number to its decimal (*base 10*) equivalent. The decimal equivalent of $0201_{(base\ 3)}$ is calculated by using 3 as the base for each place value.

$$\begin{aligned} 0201_{(base\ 3)} &= 0 \times 3^3 + 2 \times 3^2 + 0 \times 3^1 + 1 \times 3^0 \\ &= 0 + 18 + 0 + 1 \\ &= 19 \end{aligned}$$

Hence, the integer 19 represents the configuration in Fig. 2, for $N = 4$ and $K = 3$.

In general, for N cells, with K states each, there are K^N configurations, each of which can be considered as an N digit, K -ary number, and can be mapped to a decimal integer using the procedure described above. The integers will lie in the range $[0, K^N - 1]$. Conversely, a configuration can be generated from an integer in the given range, by inverting the procedure described above, converting the integer from (*base 10*) to (*base K*).

We conclude this section by emphasising the fact, that the entire space of initial configurations for a given system, with spatial extent L and dynamical parameters (τ_a, τ_p) , is *mapped one-to-one* with the range of integers $[0, K^N - 1]$, where $N = L^2$, $K = \tau_a + \tau_p + 1$. An exhaustive search of this space involves generating each possible configuration by converting the integers $[0, K^N - 1]$ from (*base 10*) to (*base K*).

B. Assumptions regarding the asymptotic behaviour of the system

As indicated in the previous subsection, this study involves systematically parsing the entire space of initial configurations for linear extent L and dynamical cycle duration K , and observing the fraction of configurations that persist, to calculate $P(\tau_a, \tau_p)$. The assumptions made about the system's asymptotic behaviour are described in this subsection.

Firstly, it is assumed that if a configuration evolves to a persistent state within $T = \Gamma$ time

steps (i.e. a configuration to which the lattice returns, after attaining a specific sequence of intermediate configurations), then the activity in the system is asymptotically persistent. This assumption is justified, as the cycle of states for each individual is deterministic and identical, and there are no stochastic effects assumed in the dynamics. The transient time Γ will depend on the duration of the cycle K as well as the linear extent L . The dependence is, quite intuitively, direct, although the exact form is not known, and is outside the scope of this study. Hence, it is assumed that a finite sized system evolves to either the persistent or the quiescent state in finite time. It is quite obvious why this assumption is important, in order to conduct an *in silico* investigation of the system under consideration (in finite time).

Secondly, a good place to begin the study would be with the simplest possible dynamics, operating on the smallest possible lattice. The smallest square lattice larger than a single cell, is four cells arranged in a square; the shortest cycle which preserves the essence of the three-phased dynamics, contains one state of each phase, i.e. singular receptive, active and passive states. So, the smallest meaningful values for the linear extent and cycle duration are $L = 2$ and $K = 3$, respectively. Now, the focus of this study is on relatively short linear extents, such small systems may be interpreted as an isolated system, or as a generating core embedded in an infinite receptive lattice. It may be noted that making the latter choice opens up further questions, such as questions about the interaction of multiple generating cores on the same lattice, and provides ways to answer them. So, in order to emulate a generating core embedded in an infinite lattice, the configurations may be padded with Λ layers of receptive cells on each boundary. Hence it is assumed that the asymptotic behaviour of a finite sized system reliably implies similar asymptotic behaviour of an infinite lattice.

The above discussion naturally leads to questions regarding the system's boundary conditions. In order to simulate an isolated patch of spatially extended media, it is intuitive to implement fixed boundary conditions, emulated by enclosing the patch within an outermost layer of perpetually passive cells [35, 36]. However, as mentioned above, to emulate a generating core configuration embedded in an infinite receptive lattice, layers of receptive cells are chosen, which are part of the system, instead of permanent passive cells which are disconnected from the internal dynamics. But even with Λ layers of receptive cells around the core configuration, a choice between fixed or periodic boundaries, remains.

For the purposes of the study, considering the length and time scales mentioned above,

the transient time at $\Gamma = 100$ steps and the padding at $\Lambda = 1$ are fixed. As confirmation, simulations for small core sizes $L = 2, K = 3$ were performed with $\Lambda = 0, 1, 2, \dots, 5$ and $\Gamma = 100, 200, \dots, 500$. It was observed that $P(\tau_a, \tau_p)$ only changes going from $\Lambda = 0$ to $\Lambda = 1$, and then attains a constant value for $\Lambda \geq 1$, and $\Gamma = 100$ was sufficient transience for the system to reach an asymptotic state, for the range of (τ_a, τ_p) explored. For the purpose of the study, periodic boundaries are maintained throughout, in order to emulate an infinite lattice as closely as possible.

C. Calculating probability of persistence $P(\tau_a, \tau_p)$ with systematic variation of parameters

As mentioned in the beginning of section III, the probability of persistence $P(\tau_a, \tau_p)$, for a given spatial extent L and active and passive phase extents (τ_a, τ_p) , is the primary quantity of interest.

As per the assumptions in the previous subsection, a transient time of $\Gamma = 100$ is defined, and the presence of active cells, i.e. $\tau_{i,j}^\Gamma \in [1, \tau_a]$ is determined in the lattice, to confirm persistence of activity. Since the observed asymptotic behaviours are qualitatively limited to two distinct classes (quiescent and persistent), satisfying this minimal condition is sufficient to confirm persistence of activity.

It is necessary to vary the phase extents τ_a and τ_p systematically, to observe how the global asymptotic phenomena of persistence or quiescence of activity is impacted by parameters which determine the dynamics at the individual level. To this end, two ways are described to vary the phase extents:

- Fixed passive - $\tau_p = P$ for a given P , i.e. the passive phase extent is anchored at P , and the variation of $P(\tau_a, \tau_p)$ is observed for increasing active phase extents τ_a .
- Fixed active - $\tau_a = A$ for a given A , i.e. the active phase extent is fixed at A , and the variation of $P(\tau_a, \tau_p)$ is observed for increasing passive phase extents τ_p .

For our study, the entire configuration space of cycle duration $K = 3, 4, \dots, 35$ is parsed, as described in previous subsections. The minimum cycle duration is $K = 3$, which corresponds to the phase extents $(\tau_a, \tau_p) \equiv (1, 1)$, and the maximum $K = 35$ corresponds to $(\tau_a, \tau_p) \equiv (17, 17)$ (as well as $(16, 18), (15, 19)$ etc. However we focus on a square patch of the (τ_a, τ_p)

parameter space, with both the τ_a and τ_p axes extending from 1 to 17, for the sake of aesthetic symmetry.)

First, the probabilities of persistence $P(\tau_a, \tau_p)$ are studied by gradually increasing the active phase extent τ_a , at the individual level, relative to a fixed passive phase extent τ_p . Next, the impact of increasing the passive phase extent τ_p is observed, on $P(\tau_a, \tau_p)$. Finally, the above observations are integrated into a heat map of the (τ_a, τ_p) parameter space, and certain insights are pointed out regarding the dependence of the global, asymptotic dynamics of the system, on the dynamical parameters at the individual level.

It should be noted that as τ_a is increased, for fixed τ_p , the overall cycle duration K simultaneously increases as well. In order to separate out the impact of a longer cycle duration K , it is important to follow up with observations of $P(\tau_a, \tau_p)$ specifically for fixed length cycles, such that $\tau_a + \tau_p = C$ for a given C . The parameters are varied with the condition that the total length of the cycle remains unchanged at $K = C + 1$. It should be emphasized that the parameter sets (τ_a, τ_p) which correspond to a given fixed cycle duration are represented on the (τ_a, τ_p) parameter space by the corresponding “anti diagonals”. By observing the dependence of $P(\tau_a, \tau_p)$ on the phase extents for a fixed cycle duration, the dependence of the probability of persistence on the overall cycle duration K , is determined.

With these methods in hand, in the following sections we present some results and observations on the probabilities of persistence of activity in the system, $P(\tau_a, \tau_p)$, as a function of the the phase extents (τ_a, τ_p) .

IV. OBSERVATIONS

We present our observations for $L = 2$. This corresponds to a 2×2 core, embedded in a 4×4 lattice as described in previous sections.

First, we take a close look at the impact of increasing the active phase extent τ_a on the probability of persistence $P(\tau_a, \tau_p)$, while keeping the passive phase extent τ_p fixed. These observations are presented in Fig. 3.

From Fig. 3, it is quite evident that increasing the active phase extent τ_a , in general, leads to a larger fraction of all possible configurations which sustain persistent activity. Specifically, for $\tau_p > 6$, a full transition, from certain quiescence to persistence of activity, occurs in $P(\tau_a, \tau_p)$, with increasing τ_a . Particularly, we highlight the sigmoidal form of

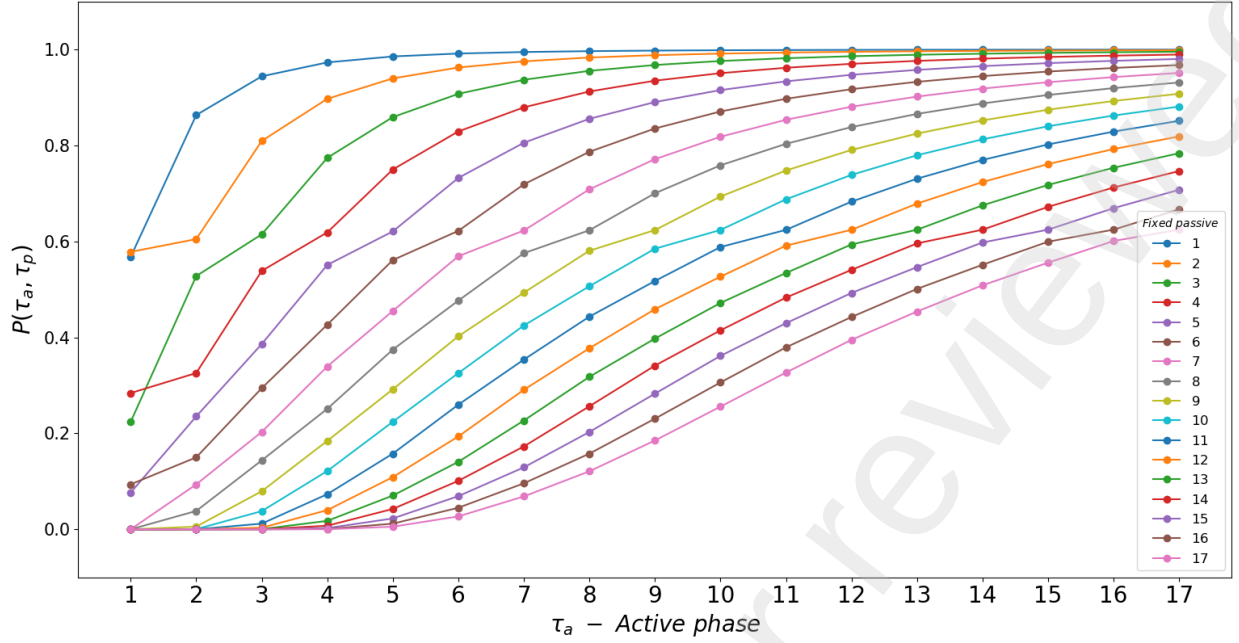


FIG. 3. Probability of persistence $P(\tau_a, \tau_p)$ for increasing active phase extent, i.e. $\tau_a \in [1, 2, \dots, 17]$ and fixed passive phase extent τ_p . Note the inflection point in each graph at $\tau_a = \tau_p$.

this transition. A slight flattening of the slope of the curve around the middle is observed, discernible for the curves corresponding to $\tau_p \geq 5$. These *inflection points* correspond to the parameter sets (τ_a, τ_p) where $\tau_a = \tau_p$. The ratio $\rho = \frac{\tau_a}{\tau_p}$ switches from $\rho \leq 1$ to $\rho \geq 1$ at these inflection points, which correlates to a change in the concavity of the curve.

As a natural subsequent aspect, the consequence of increasing the passive phase extent τ_p is investigated, for fixed τ_a , on the probability of persistent activity $P(\tau_a, \tau_p)$. These observations are shown in Fig. 4.

From Fig. 4, it is found that increasing τ_p , for a given τ_a , generally has a diminishing effect on $P(\tau_a, \tau_p)$, which is consistent with previous observations. The form of the transition may be described as “anti sigmoidal”, and the points where $\tau_a = \tau_p$ correspond to inflection points, consistent with previous observations. Since there are no theoretical upper bounds to τ_a or τ_p , it may be concluded that given any τ_a , the fixed point (which corresponds to global quiescence) becomes the stable outcome for a sufficiently long passive phase extent τ_p .

In order to integrate the previous observations into a consolidated picture of the overall dependence of $P(\tau_a, \tau_p)$ on τ_a and τ_p , the (τ_a, τ_p) parameter space plot is presented in Fig. 5.

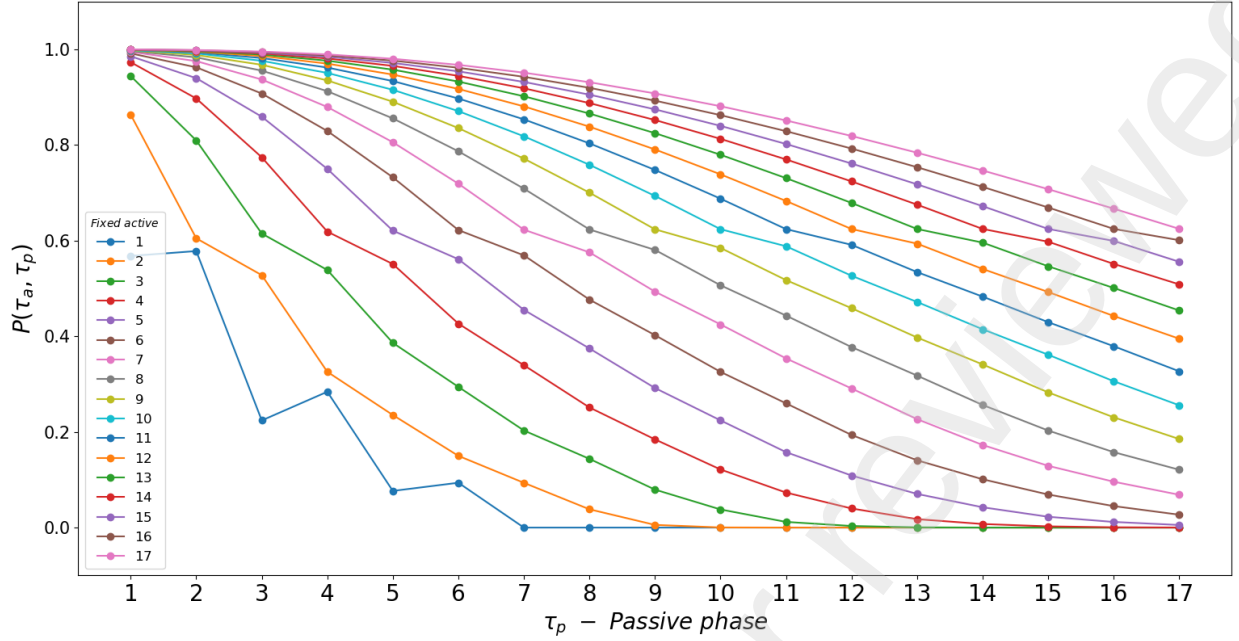


FIG. 4. Probability of persistence $P(\tau_a, \tau_p)$ for increasing passive phase extent, i.e. $\tau_p \in [1, 2, \dots, 17]$ and fixed active phase extent τ_a .

To contextualise Fig. 5 with previous observations, it should be noted that increasing active and passive phase extents (Figs. 3 and 4) correspond to horizontal and vertical slices of the parameter space, respectively. As mentioned earlier, the parameter sets (τ_a, τ_p) such that $\tau_a + \tau_p = C$ constitute the anti diagonals of the parameter space. For an analysis of the case of fixed length cycles, the anti diagonals beginning from the point $(\tau_a, \tau_p) = (1, 1)$ in Fig. 5 are explored.

For example, for the fixed cycle duration $K = 3$, $(\tau_a, \tau_p) \equiv (1, 1)$ is the only possible pair of parameter values. For $K = 4$, $(\tau_a, \tau_p) \equiv (1, 2)$ and $(2, 1)$, and so on, up to $K = 19$, which is the longest anti diagonal in Fig. 5, beginning from $(\tau_a, \tau_p) = (1, 17)$ to $(\tau_a, \tau_p) = (17, 1)$. The longest anti diagonal of the parameter space, corresponding to a fixed cycle duration of $K = 19$ is presented in Fig. 6.

From Fig. 6 it is evident that, even when the active phase extent τ_a increases along with a simultaneous reduction of the passive phase extent τ_p , such that the overall cycle duration of the individual dynamics remains constant $\tau_a + \tau_p = C$, $K = C + 1$, the probability of persistence $P(\tau_a, \tau_p)$ increases as a consequence. This transition also has the characteristic sigmoidal shape, along with a similar inflection point at $\tau_a = \tau_p$, noted in earlier contexts.

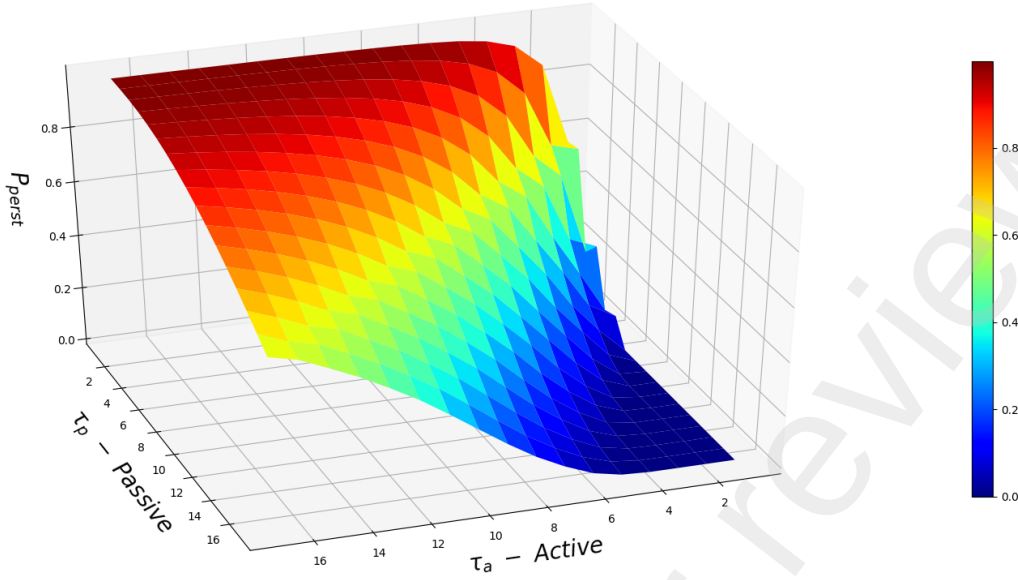


FIG. 5. Probability of persistence $P(\tau_a, \tau_p)$ in the (τ_a, τ_p) parameter space. Observe the regions of certainty, for high τ_a and low τ_p , for which $P(\tau_a, \tau_p) \approx 1$, and similarly for low τ_a and high τ_p , for which $P(\tau_a, \tau_p) \approx 0$. Colours indicate $P(\tau_a, \tau_p)$.

The “regions of certainty” in parameter space (Fig. 5) are discernible near the horizontal (active) axis, where $P(\tau_a, \tau_p) \approx 1$, and the vertical (passive) axis, where $P(\tau_a, \tau_p) \approx 0$. These regions, marked by contour lines in Fig. 7, emerge on either side of the main diagonal, which corresponds to parameter sets (τ_a, τ_p) such that $\tau_a = \tau_p$. Correspondingly, the middle region may be considered as a “region of ambiguity”, where $0 < P(\tau_a, \tau_p) < 1$.

The slopes of the contour lines contain information about the “rate of growth” of these regions, and can be used to calculate the probabilities of persistence for longer cycle duration K . For instance, given a parameter set (τ_a, τ_p) , it is possible to locate the corresponding point in the (extended) parameter space, and extend the contour lines and observe the region in which the point is located, thereby determining the probability of persistence associated with the given parameter set.

The relatively constant slope of the contour lines, across the range of probabilities in the region of ambiguity $P(\tau_a, \tau_p) \in (0, 1)$, indicate uniform growth of each constituent sub-region, as the active and passive phase extents increase. In other words, as the parameters τ_a and τ_p are increased, and a specific range of probabilities of persistence is considered, say for instance $P(\tau_a, \tau_p) \in (0.4, 0.5)$, then the fraction of persistent configurations in this range

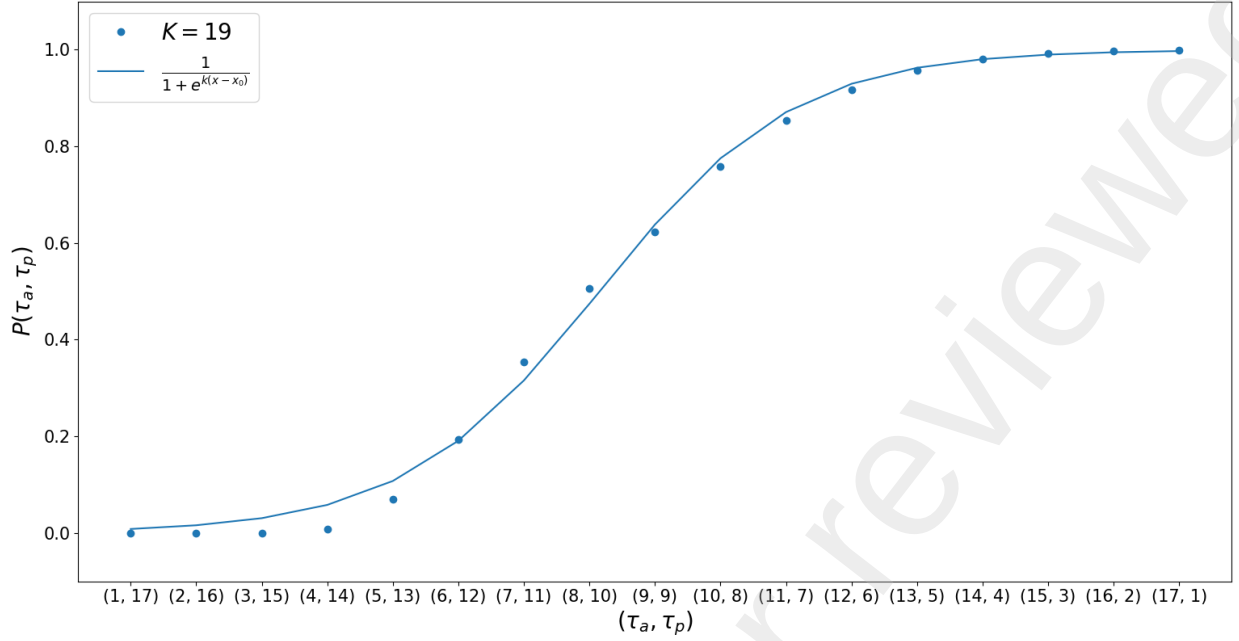


FIG. 6. $P(\tau_a, \tau_p)$ w.r.t increasing τ_a and simultaneously decreasing τ_p , for fixed cycle duration $K = 19$. The (τ_a, τ_p) pairs are ordered for each K , such that the active phase extent τ_a increases from left to right, and the passive phase extent τ_p decreases from left to right. The data is fitted to the standard form of a sigmoid (mentioned in the legend), with parameters $x_0 = 8.15, k = 0.67$ increase equivalently to other ranges of probabilities of persistence, say $P(\tau_a, \tau_p) \in (0.7, 0.8)$.

V. SUMMARY AND DISCUSSION

A brief summary is presented, highlighting the results and some possible future directions of research.

A cellular automata model, in which space, time and states have a discrete representation, is considered. The dynamics at the individual level is cyclic and excitable, consisting of three phases — receptive, active and passive. We investigate the influence of the temporal properties of the dynamics at the individual level, on the global asymptotic state of the system. To this end, first a scheme to uniquely identify each possible configuration of a system with a given lattice size $N = L \times L$, and specified active and passive phase extents (τ_a, τ_p) , is described. Certain characteristic trends of the probability function $P(\tau_a, \tau_p)$ with respect to systematic variations of τ_a and τ_p are then discerned. The main observations are:

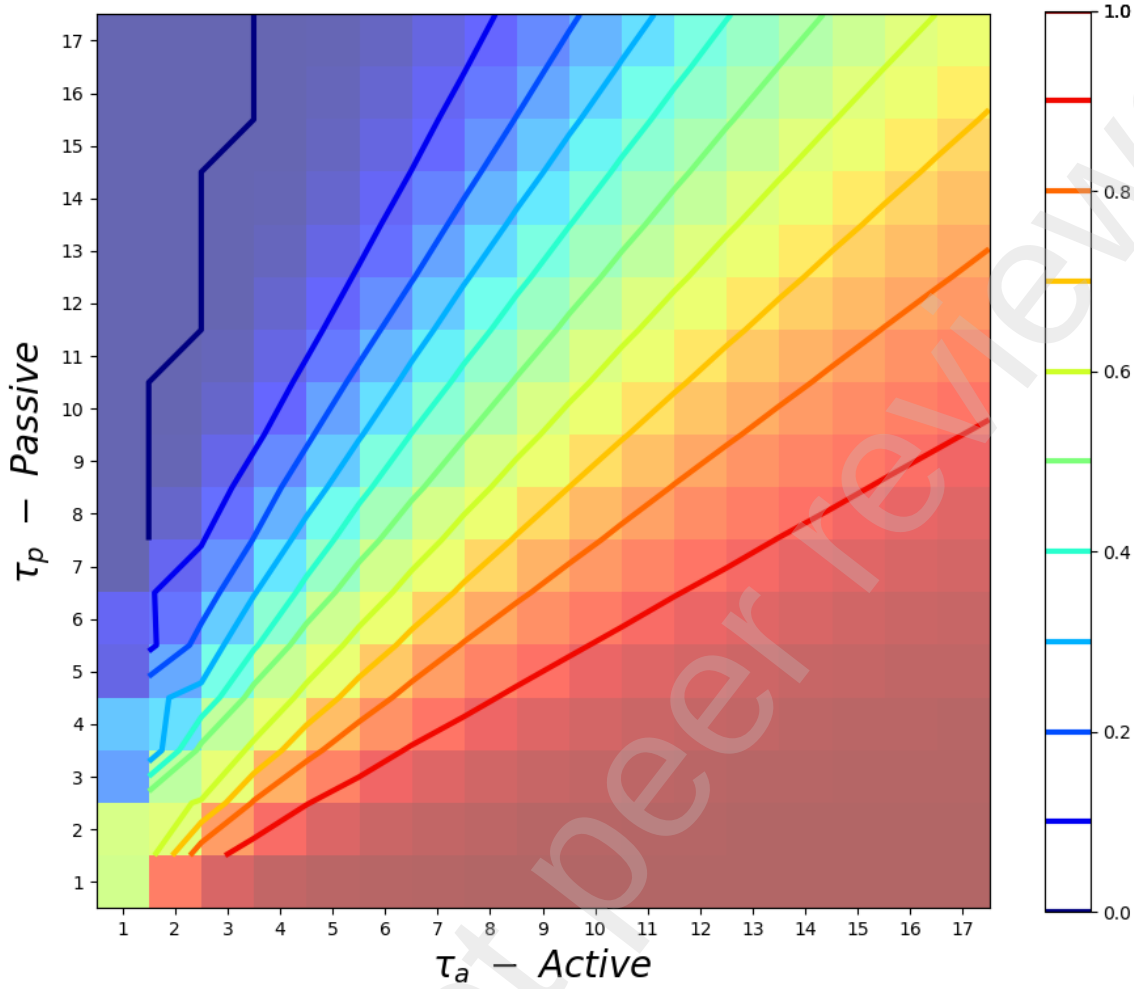


FIG. 7. (τ_a, τ_p) parameter space with contour lines marking regions of similar probabilities of persistence.

- Increasing active phase extent τ_a enhances probability of persistence $P(\tau_a, \tau_p)$ (Fig. 3).
- Increasing the passive phase extent τ_p diminishes the probability of persistence $P(\tau_a, \tau_p)$ (Fig. 4).
- The transition of the probability of persistence $P(\tau_a, \tau_p)$ with respect to τ_a (τ_p) has a characteristic sigmoidal (anti sigmoidal) shape.
- There exists an inflection point at $\tau_a = \tau_p$, where the concavity of the probability curve changes.
- The above conclusions are true even as the parameter changes occur simultaneously while maintaining the overall cycle duration K (Fig. 6).

- In the (τ_a, τ_p) parameter space (Fig. 5), there are “regions of certainty”, where $P(\tau_a, \tau_p) \approx 0$ for longer passive phase extents (large τ_p), or $P(\tau_a, \tau_p) \approx 1$ for longer active phase extents (large τ_a).
- The rate of growth of sub-regions of equal probability (w.r.t increasing parameter values), on the (τ_a, τ_p) parameter space, is approximately equivalent, as conveyed by the slopes of the contour lines which form the boundaries of these sub-regions (Fig. 7).

The primary motivation for the study, as discussed in Section I, has been to understand the impact of the proportion of active and passive phases, which are local timescales of the dynamics of each cell, on the probability of emergent, sustained oscillations in an excitable system. There are questions along several directions which arise consequently. We consider some of them, specifically along analytical and computational lines. Finally, we discuss some possible directions to broaden the scope of the model, inspired by real-world phenomena.

Analytically, it would be interesting to explore the scaling of the probability of persistence as the size of the lattice increases. Specifically, consider the Greenberg-Hastings dynamics with the most general state space $S = \{0, 1, \dots, \tau_a, \tau_a + 1, \dots, \tau_a + \tau_p\}$, $|S| = K$ as a transformation T on configurations of a finite 2D lattice of length L , represented as $B_L = \{0, \dots, L-1\}^2$, with $N = L^2$ cells. Let the (finite) set of configurations be represented as $X = S^{B_L}$. We know that the number of configurations in X is $|X| = K^N$. An attempt may be made to enumerate the number of persistent configurations. One approach could be to firstly consider the set of eventual images $Y = \bigcap_{n=1}^{\infty} T^n X$. Y consists of the asymptotic configurations of the system, namely configurations which are persistent and part of a limit cycle, as well as the extinct configuration (where each cell is at the 0 state). If $|Y| = y$, i.e. there are y number of configurations in the set of asymptotic configurations, then it may be concluded that the number of persistent configurations is at least $y - 1$. In addition, there are a significant number of so called “garden of eden” configurations [13], which are configurations that cannot be attained by applying the transformation T on any other configuration, and as such can only be considered as initial configurations, which may be persistent. All the configurations which form the intermediate path between the persistent garden of eden configurations and it’s attractor, as they evolve towards an eventual image, would also be counted as persistent. The sum total of such configurations would enable the calculation of the probability of persistence $P(N, \tau_a, \tau_p)$. Of course, as can be readily

concluded from the results presented in this study, the number of persistent configurations would vary with τ_a and τ_p for the same N . The form of this dependence has been presented here, and perhaps provides a hint towards future development of such scaling relationships.

The challenging problem of inferring the structure of a network, from the dynamics occurring at their nodes, has been successfully approached computationally [38], using a non parametric Bayesian method, aided by Markov Chain Monte Carlo sampling. It is heartening to learn about the utility of simple models, such as the SIS model, which describe dynamics similar to the Greenberg-Hastings model, for such complex tasks. Extending the argument regarding structural cycles presented in Section I, it would be interesting to explore if the inference of structural features can be enhanced by further consideration of the implications of emergent phenomena such as persistence.

In the context of real-world systems, the restrictive scenario of cells with identical properties, operating in a deterministic manner, could be relaxed, to obtain a much wider gamut of patterns. Specifically, lattices with heterogeneous local properties, where each cell has a unique set of parameters, could be explored. The parameters which could be diversified include radius of interaction, active and passive phase extents, as well as update thresholds [25]. As mentioned earlier, similar dynamics has also been considered with probabilistic updates [29, 30] in the context of infectious diseases on networks, which broadens the scope of the dynamical model beyond lattices to graphs or networks [6]. The question of persistence could be further explored with respect to such generalizations.

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VII. DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

VIII. AUTHOR DECLARATIONS

A. Conflict of interest

The authors have no conflicts of interest to disclose.

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